

Discrete Structures

Lecture # 4

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Recap

- Conditional Statements
- Bi-conditional Statements
- Conversion of NLP to Argument and vice versa
- Inverse / Converse / Contrapositive of statements
- Examples / Exercise

Argument

An **argument** is a list of statement called **premises** (or **assumptions** or **hypotheses**) followed by a statement called the **conclusion**.

Valid and Invalid Arguments

- ❑ Propositional logic can be used as a math model to investigate the validity of arguments.
- ❑ An argument is a sequence of statements.
- ❑ All but the final statements are called premises.
- ❑ Final statement is called conclusion.
- ❑ Valid Argument: If the premises are all true then the conclusion is also true.

$$\square P_1 \wedge \dots \wedge P_n \vdash Q$$

i.e. Premises logically implies the conclusion.

Argument Validity

- Two Ways:

- Using truth Tables

- Reason at a higher level using generally valid rules (inference values).

Argument

P1 Premise

P2 Premise

P3 Premise

P4 Premise

.....

.....



$\therefore C$

Conclusion

Valid Argument

An argument is **valid** if the **conclusion** is **true** when all the **premises** are **true**.

Invalid Argument

An argument is **invalid** if the **conclusion** is **false** when all the premises are **true**.

Example of Valid Argument

Show that the following argument form is valid:

$p \rightarrow q$	premise
p	premise
$\therefore q$	Conclusion

Premise			Conclusion	
p	q	$p \rightarrow q$	p	q
T	T	T	T	T
T	F	F	T	F
F	T	T	F	T
F	F	T	F	F

The given argument is valid.

Example of Invalid Argument

Show that the following argument form is valid:

$p \rightarrow q$ premise
 q premise
 $\therefore p$ Conclusion

Premise			Conclusion	
p	q	$p \rightarrow q$	q	p
T	T	T	T	T
T	F	F	F	T
F	T	T	T	F
F	F	T	F	F

The given argument is Invalid.

Example

Show that the following argument form is valid:

$p \vee q$ premise

Premise

$p \rightarrow \sim q$ premise

Conclusion

$p \rightarrow r$ premise

$\therefore r$ Conclusion

p	q	r	$p \vee q$	$p \rightarrow \sim q$	$p \rightarrow r$	r
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Example

Premise

Conclusion

p	q	r	$p \vee q$	$p \rightarrow \sim q$	$p \rightarrow r$	r
T	T	T	T	F	T	T
T	T	F	T	F	F	F
T	F	T	T	T	T	T
T	F	F	T	T	F	F
F	T	T	T	T	T	T
F	T	F	T	T	T	F
F	F	T	F	T	T	T
F	F	F	F	T	T	F

The given argument is invalid.

Exercise – 1

If Tariq is not on team A, **then** Hameed is on team B

If Hameed is not on team B, **then** Tariq is on team A.

∴ If Hameed is **not** on team B, **then** Tariq is not on team A.

Solution:

Let t = Tariq is on team A

h = Hameed is on team B

Exercise – 1 – Cont..

Solution: Let

t = Tariq is on team A

h = Hameed is on team B

1. If Tariq is not on team A, then Hameed is on team B -- $(\sim t \rightarrow h)$
2. If Hameed is not on team B, then Tariq is on team A -- $(\sim h \rightarrow t)$
3. $\therefore$ If Hameed is not on team A, then Tariq is not on team B
 $B \text{ -- } \sim h \rightarrow \sim t$

Exercise – 1 – Cont..

1. $(\sim t \rightarrow h)$
2. $(\sim h \rightarrow t)$
3. $\therefore \sim h \rightarrow \sim t$

		Premise		Conclusion
t	h	$\sim t \rightarrow h$	$\sim h \rightarrow t$	$\sim h \rightarrow \sim t$
T	T	T	T	T
T	F	T	T	F
F	T	T	T	T
F	F	F	F	T

The given argument is invalid.

Exercise – 2

If at least one of these two numbers is divisible by 6,
then the product of these two numbers is divisible by 6.

Neither of these two numbers is divisible by 6.

∴ The product of these two numbers is not divisible by 6.

Solution: Let

d = at least one of these two numbers is divisible by 6

p = product of these two numbers is divisible by 6

Exercise – 2 – Cont..

Solution: Let,

d = at least one of these two numbers is divisible by 6

p = product of these two numbers is divisible by 6.

1. **If** at least one of these two numbers is divisible by 6,
then the product of these two numbers is divisible by 6
-- $(d \rightarrow p)$
2. **Neither** of these two numbers is divisible by 6 -- $\sim d$
3. $\therefore$ The product of these two numbers is **not** divisible by 6 -- $\sim p$

Exercise – 2 – Cont..

Solution:

1. $(d \rightarrow p)$
2. $\sim d$
3. $\therefore \sim p$

		Premise	Conclusion	
d	p	$d \rightarrow p$	$\sim d$	$\sim p$
T	T	T	F	F
T	F	F	F	T
F	T	T	T	F
F	F	T	T	T

The given argument is invalid.

Exercise – 3

If I got an Eid Bonus, I'll buy a stereo.

If I sell my motorcycle, I'll buy a stereo.

∴ If I get Eid bonus or I sell my motorcycle, then I'll buy a stereo.

Solution: Let

e = I got an Eid Bonus

s = I'll buy a stereo

m = I sell my motorcycle

Exercise – 3 – Cont..

Solution: Let, e = I got an Eid Bonus; s = I'll buy a stereo; m = I sell my motorcycle

1. If I got an Eid Bonus, I'll buy a stereo -- $(e \rightarrow s)$
2. If I sell my motorcycle, I'll buy a stereo -- $(m \rightarrow s)$
3. $\therefore$ If I get Eid bonus or I sell my motorcycle, then I'll buy a stereo – $e \vee m \rightarrow s$

e	s	m	$e \rightarrow s$	$m \rightarrow s$	$e \vee m$	$e \vee m \rightarrow s$
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Exercise – 3 – Cont..

Solution: $(e \rightarrow s); (m \rightarrow s); \therefore e \vee m \rightarrow s$

e	s	m	$e \rightarrow s$	$m \rightarrow s$	$e \vee m$	$e \vee m \rightarrow s$
T	T	T	T	T	T	T
T	T	F	T	T	T	T
T	F	T	F	T	T	F
T	F	F	F	T	T	F
F	T	T	T	T	T	T
F	T	F	T	T	F	T
F	F	T	T	F	T	F
F	F	F	T	T	F	T

The given argument is valid.

Inference Rule

- ❑ To help showing that a conclusion follows logically from a set of premises we may apply inference rules on the form,

$$p_1 \dots p_n \vdash q$$

- ❑ The validity of the rule is ensured

If $(p_1 \wedge \dots \wedge p_n) \rightarrow q$ is a Tautology

- A tautology is a statement which is always true. E.g. $p \vee \sim p$.

Inference Rule

□ Modus Ponens

$$\frac{p \quad p \rightarrow q}{\therefore q} \text{ (Based on } [p \wedge (p \rightarrow q) \rightarrow q])$$

□ Modus Tollens

$$\frac{p \rightarrow q \quad \sim q}{\therefore \sim p} \text{ (Based on } [(p \rightarrow q) \wedge \sim q \rightarrow \sim p])$$

□ Generalization

$$\frac{p}{\therefore p \vee q}, \frac{q}{\therefore p \vee q}$$

Inference Rule

□ Specialization

$$\frac{p \wedge q}{\therefore p}, \frac{p \wedge q}{\therefore q}$$

□ Elimination

$$\frac{p \vee q \quad \sim q}{\therefore p}, \frac{p \vee q \quad \sim p}{\therefore q}$$

□ Conjunction

$$\frac{p \quad q}{\therefore p \wedge q}$$

□ Transitivity

$$\frac{p \rightarrow q \quad q \rightarrow r}{\therefore p \rightarrow r}$$

Inference Rule -- Application – An Example

❑ Example: You are about to leave for University in the morning and discover that you don't have your glasses. You know the following statements are true.

- A. If I was reading the newspaper in the kitchen, then my glasses are on the kitchen table.
- B. If my glasses are on the kitchen table, then I saw them at breakfast.
- C. I did not see my glasses at breakfast.
- D. I was reading the newspaper in the living room or I was reading the newspaper in the kitchen.
- E. If I was reading the newspaper in the living room then my glasses are on the coffee table.

Where are the glasses??

Inference Rule -- Application – An Example

Assume, RK= Reading the newspaper in the kitchen.

GK= Glasses are on the kitchen table.

SB= I saw my glasses at breakfast.

RL= Reading the newspaper in the living room.

GC= Glasses are on the coffee table.

So by rules of inference,

$$1. \frac{\begin{array}{l} RK \rightarrow GK \quad (\text{by A}) \\ GK \rightarrow SB \quad (\text{by D}) \end{array}}{\therefore RK \rightarrow SB \text{ (Transitivity)}}$$

$$2. \frac{\begin{array}{l} RK \rightarrow SB \quad (\text{by 1}) \\ \sim SB \quad (\text{by C}) \end{array}}{\therefore \sim RK \text{ (by modus tollens)}}$$

$$3. \frac{\begin{array}{l} RL \vee RK \quad (\text{by D}) \\ \sim RK \quad (\text{by 2}) \end{array}}{\therefore RL \text{ (by elimination)}}$$

$$4. \frac{\begin{array}{l} RL \rightarrow GC \quad (\text{by C}) \\ RL \quad (\text{by 3}) \end{array}}{\therefore GC \text{ (by modus ponens)}}$$

So the Glasses are on the Coffee table.

Contradiction and Valid Arguments.

□ Contradiction Rule

Suppose p is some statement whose truth you wish to deduce.

If you can show that the supposition that p is false leads logically to a contradiction, then you can conclude that p is true.

Contradiction and Valid Arguments.

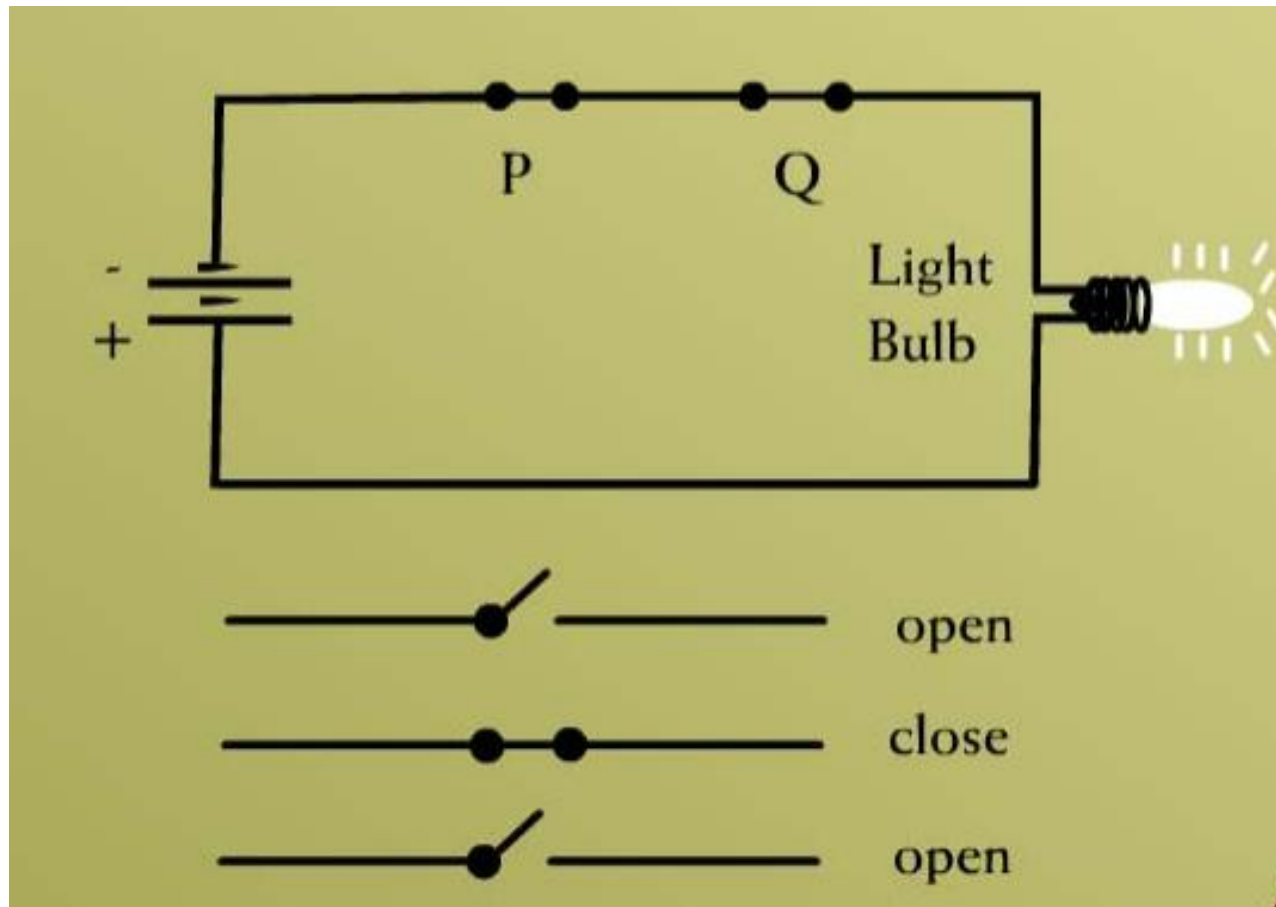
□ Contradiction Rule

$\frac{\sim p \rightarrow c}{\therefore p}$, where c is a contradiction

p	$\sim p$	c	$\sim p \rightarrow c$	p
T	F	F	T	T
F	T	F	F	F

Logical heart of the method of proof by contradiction.

Switches in Series



Switches in Series

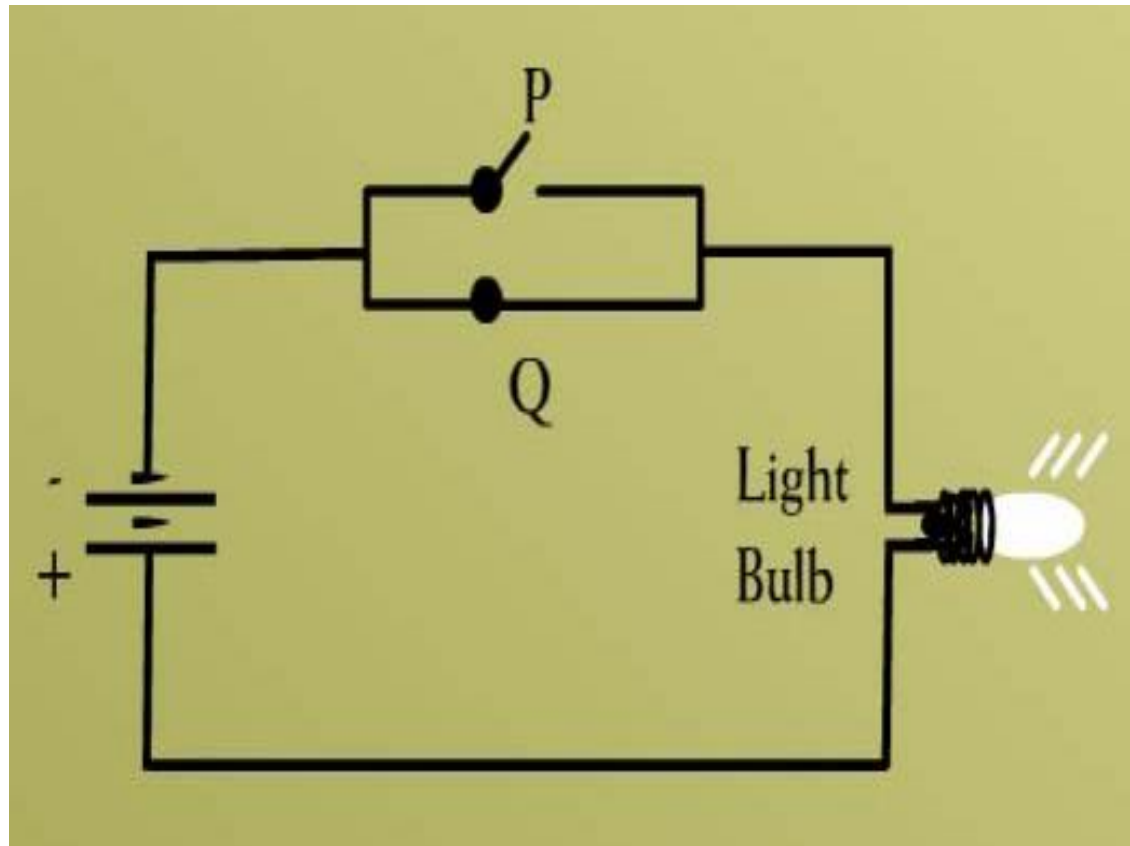
Switches		Light Bulb
P	Q	State
Open	Open	Off
Open	Closed	Off
Closed	Open	Off
Closed	Closed	On

Switches in Series

Switches		Light Bulb
P	Q	State
T	T	T
T	F	F
F	T	F
F	F	F

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

Switches in Parallel



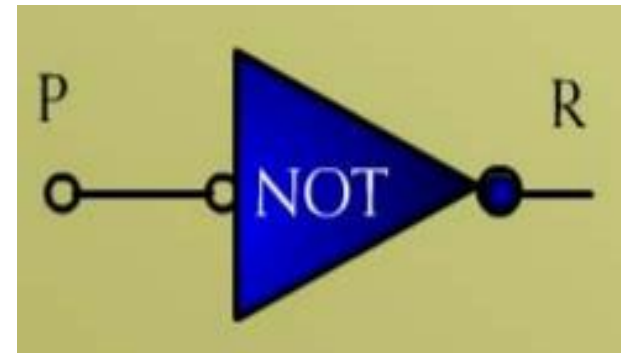
Switches in Parallel

Switches		Light Bulb
P	Q	State
T	T	T
T	F	T
F	T	T
F	F	F

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

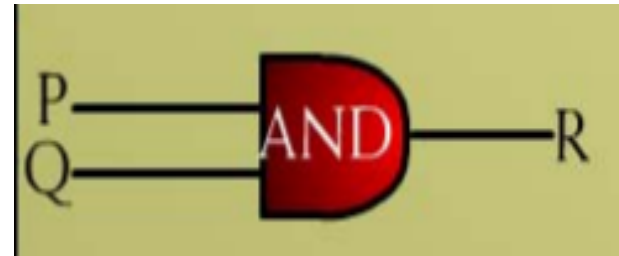
Not Gate or Inverter

Input	Output
P	Q
1	0
0	1



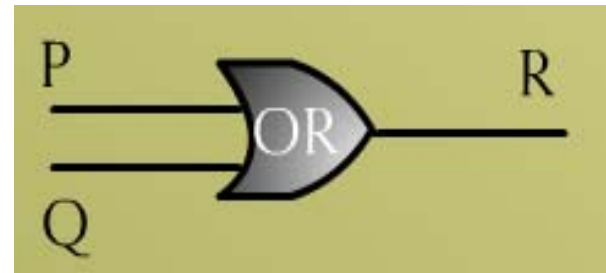
AND Gate

Input		Output
P	Q	R
1	1	1
1	0	0
0	1	0
0	0	0

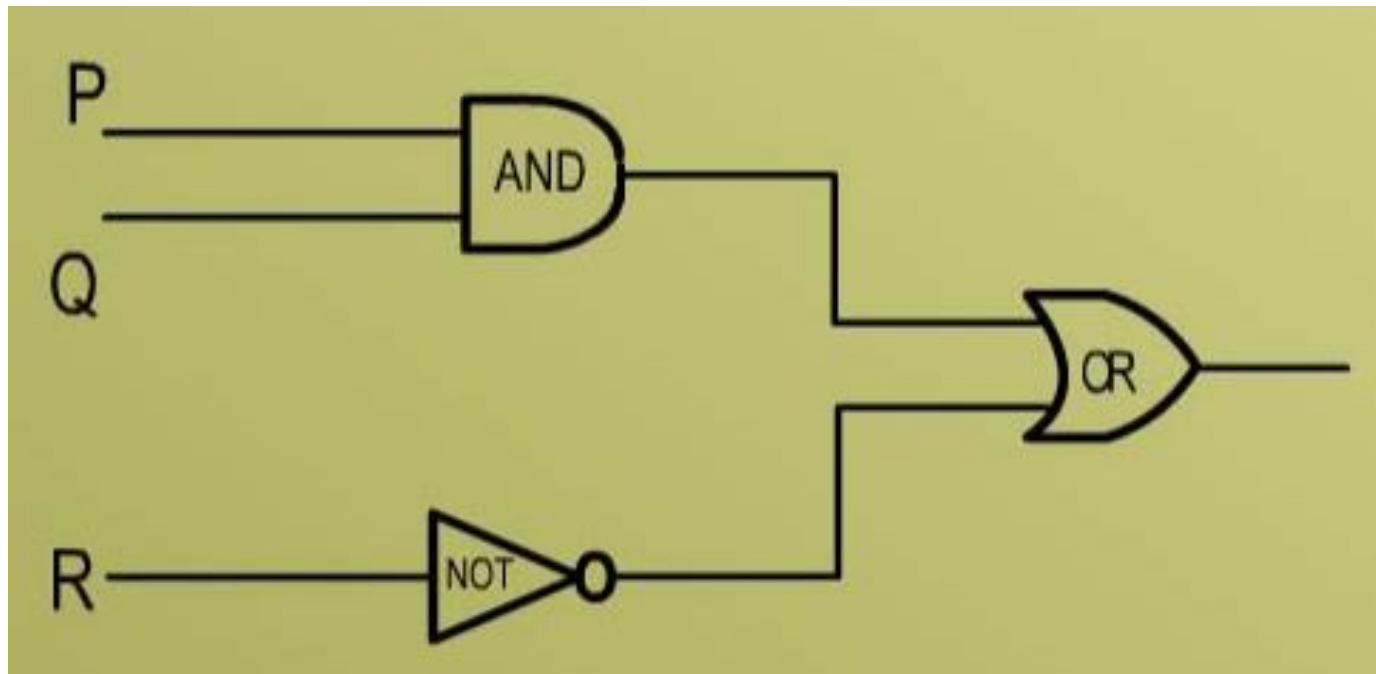


OR Gate

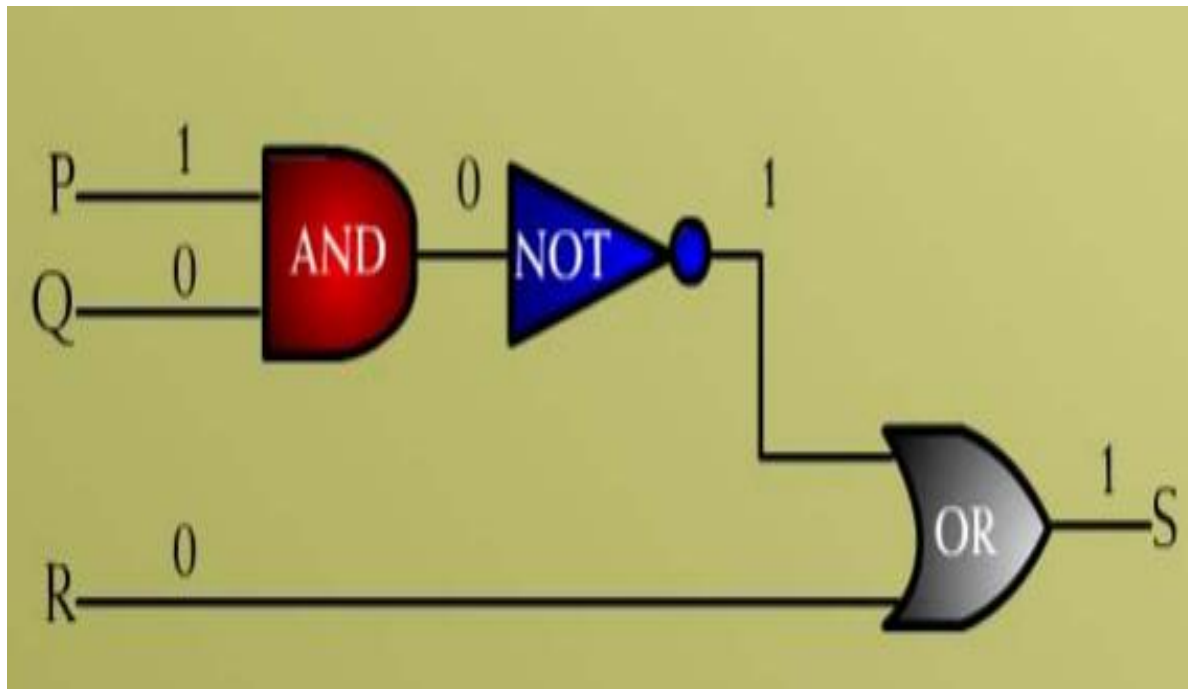
Input		Output
P	Q	R
1	1	1
1	0	1
0	1	1
0	0	0



Combinational Circuit



Output for a given Input



Input / Output table for a circuit

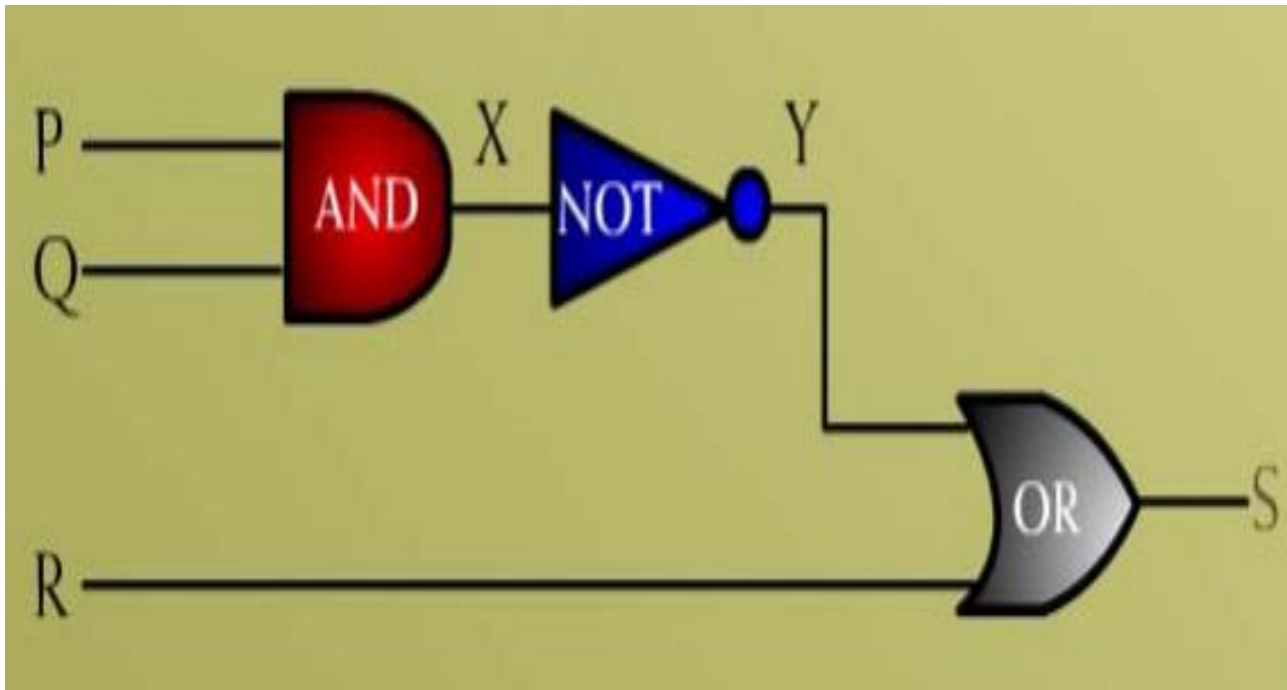


Table for a circuit

P	Q	R	X	Y	S
1	1	1			
1	1	0			
1	0	1			
1	0	0			
0	1	1			
0	1	0			
0	0	1			
0	0	0			

Table for a circuit – Cont.

P	Q	R	X	Y	S
1	1		1		
1	1		1		
1	0		0		
1	0		0		
0	1		0		
0	1		0		
0	0		0		
0	0		0		

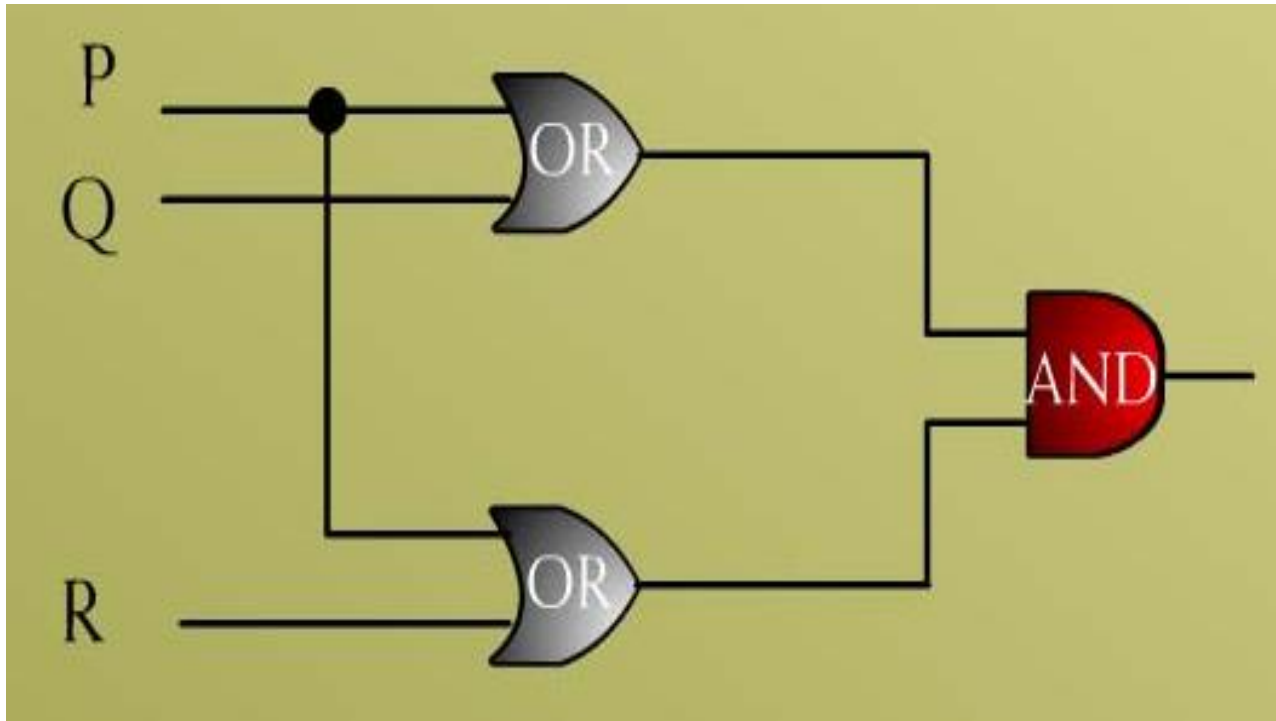
Table for a circuit – Cont.

[illegible]

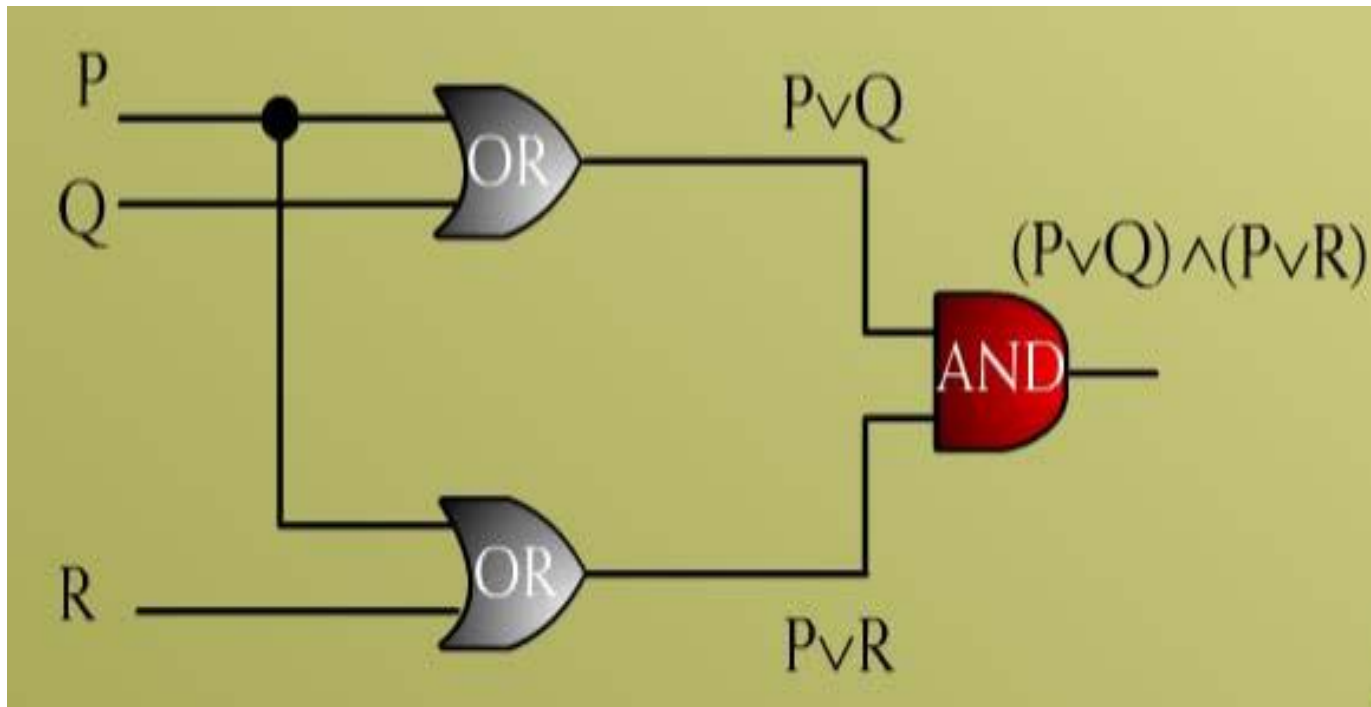
Table for a circuit – Cont.

P	Q	R	X	Y	S
		1		0	1
		0		0	0
		1		1	1
		0		1	1
		1		1	1
		0		1	1
		1		1	1
		0		1	1

Boolean Expression for a Circuit

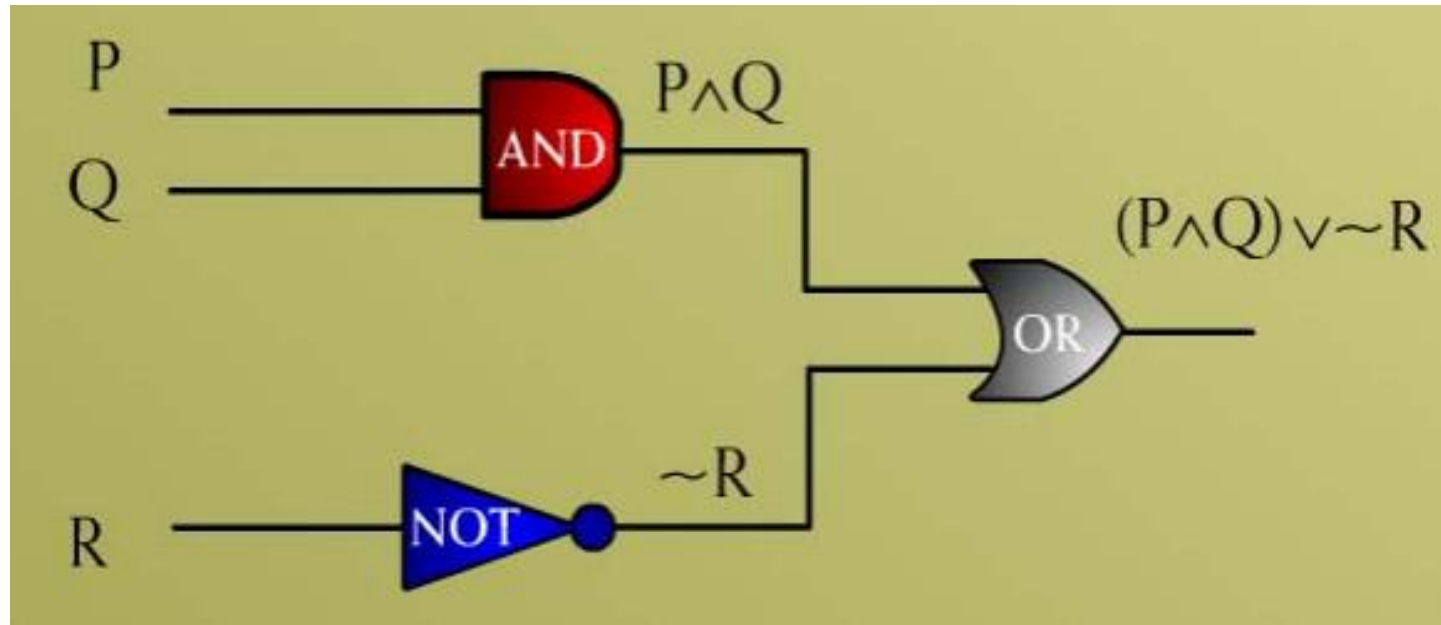


Boolean Expression for a Circuit



Circuit for a Boolean Expression

$$(P \wedge Q) \vee \sim R$$



Circuit for Input / Output Table

INPUTS			OUTPUTS
P	Q	R	S
1	1	1	0
1	1	0	1
1	0	1	0
1	0	0	0
0	1	1	1
0	1	0	0
0	0	1	0
0	0	0	0

Circuit for Input / Output Table – Sol.

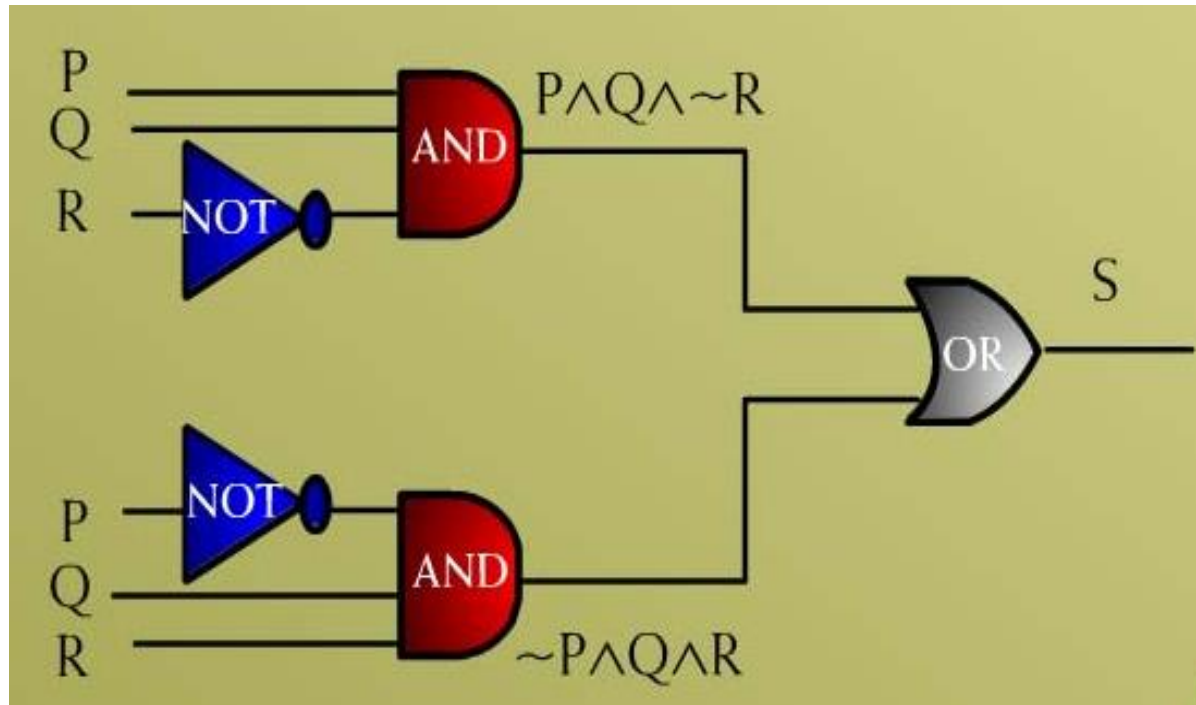
INPUTS			OUTPUTS
P	Q	R	S
1	1	1	0
1	1	0	1
1	0	1	0
1	0	0	0
0	1	1	1
0	1	0	0
0	0	1	0
0	0	0	0

$$P \wedge Q \wedge \sim R$$


$$\sim P \wedge Q \wedge R$$


Circuit Diagram

$$(P \wedge Q \wedge \sim R) \vee (\sim P \wedge Q \wedge R) = S$$



Exercise – 1

Design a circuit to take **input** signals **P**, **Q**, and **R** and **output** a **1** if, and only if, **P** and **Q** have the same value and **Q** and **R** have opposite values.

Exercise – 1: Sol.

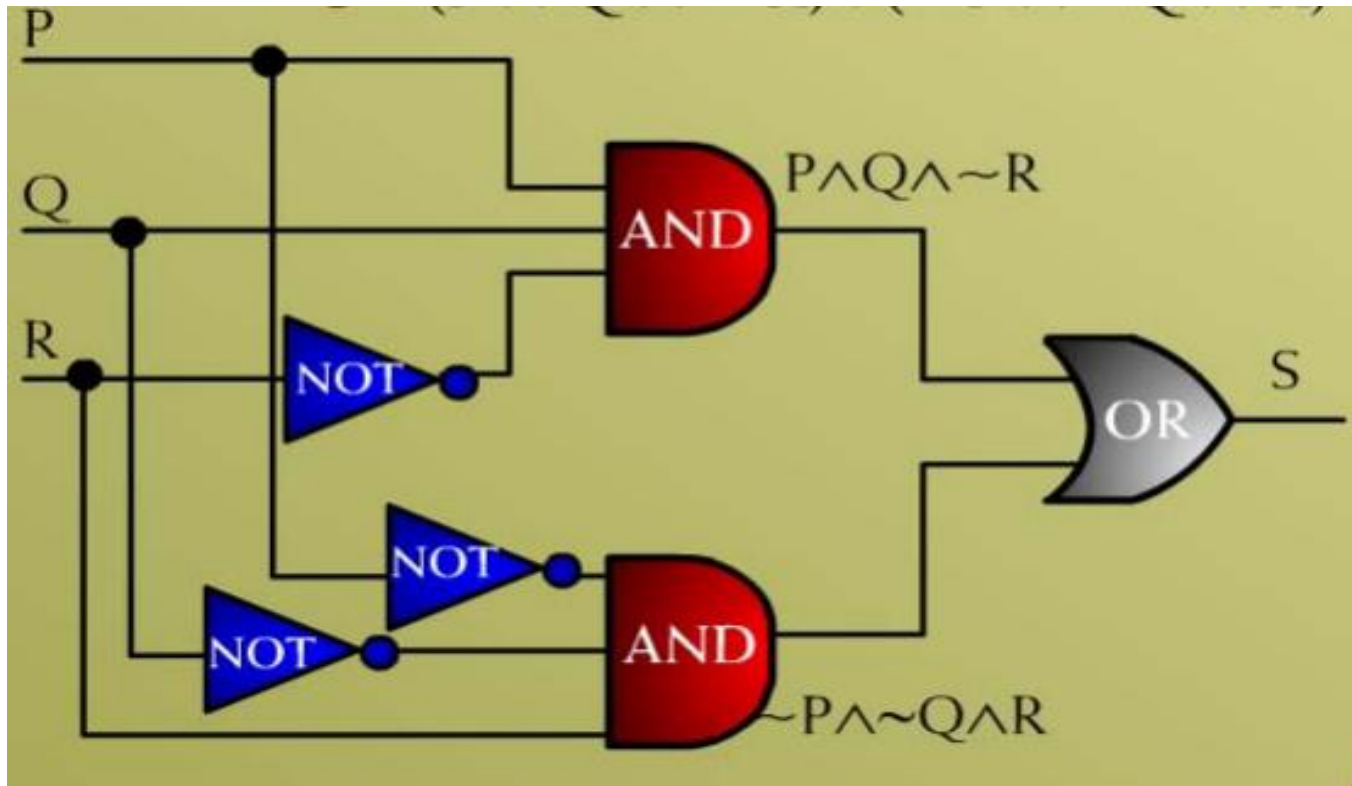
INPUTS			OUTPUTS
P	Q	R	S
1	1	1	0
1	1	0	1
1	0	1	0
1	0	0	0
0	1	1	0
0	1	0	0
0	0	1	1
0	0	0	0

$$P \wedge Q \wedge \sim R$$

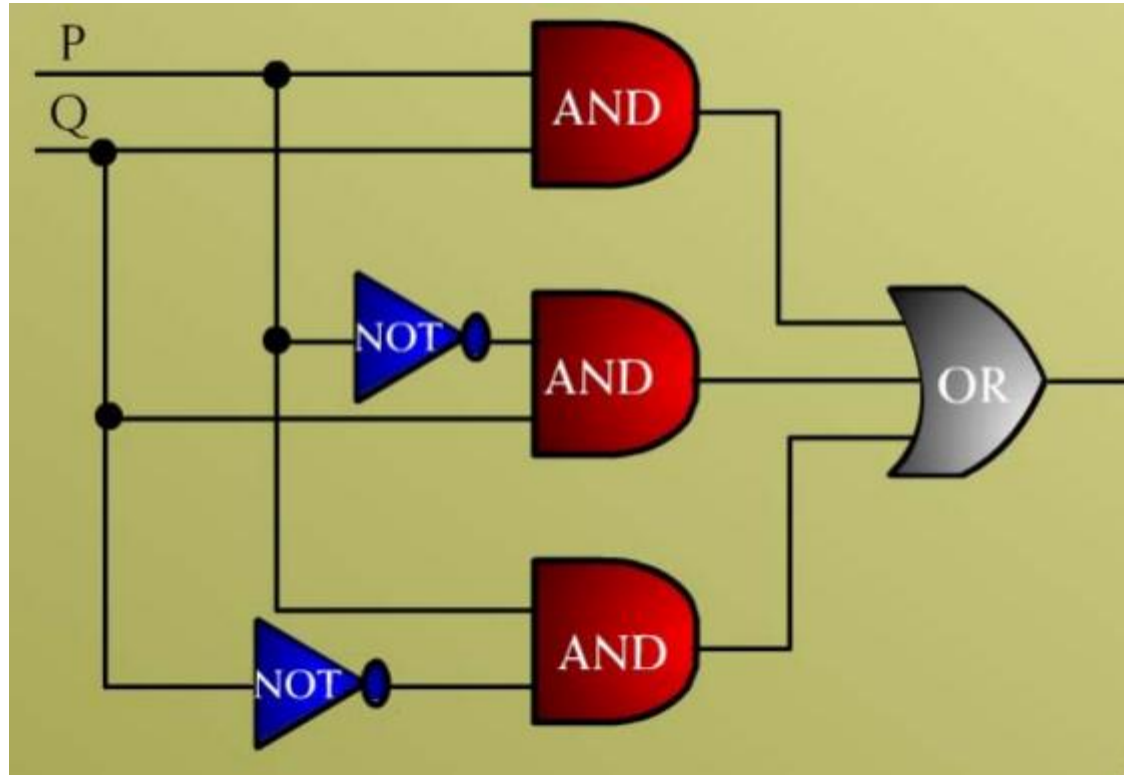
$$\sim P \wedge \sim Q \wedge R$$

Exercise – 1: Sol.

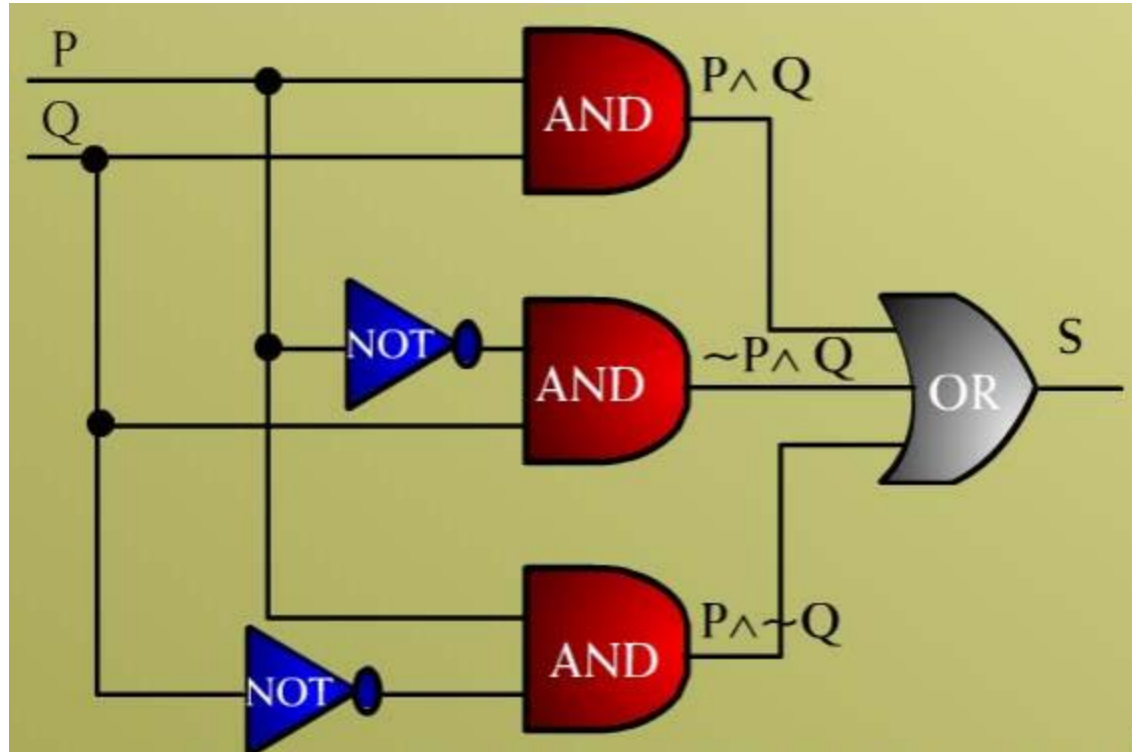
$$S = (P \wedge Q \wedge \sim R) \vee (\sim P \wedge \sim Q \wedge R)$$



Exercise – 2



Exercise – 2 : Sol.



OUTPUT:

$$S = (P \wedge Q) \vee (\neg P \wedge Q) \vee (P \wedge \neg Q)$$

Exercise – 2 : Sol.

Statement	Reason
$(P \wedge Q) \vee (\sim P \wedge Q) \vee (P \wedge \sim Q)$	
$\equiv (P \wedge Q) \vee (\sim P \wedge Q) \vee (P \wedge \sim Q)$	
$\equiv (P \vee \sim P) \wedge Q \vee (P \wedge \sim Q)$	Distributive law
$\equiv t \wedge Q \vee (P \wedge \sim Q)$	Negation law
$\equiv Q \vee (P \wedge \sim Q)$	Identity law
$\equiv (Q \vee P) \wedge (Q \vee \sim Q)$	Distributive law

Exercise – 2 : Sol.

Statement	Reason
$\equiv (Q \vee P) \wedge t$	Negation law
$\equiv Q \vee P$	Identity law
$\equiv Q \vee P$	Commutative law

Thus $(P \wedge Q) \vee (\sim P \wedge Q) \vee (P \wedge \sim Q) \equiv P \vee Q$